

知识点一：求矩阵的秩，做法是“消消乐”

B站：猴博士爱讲课
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消消乐的目标：将矩阵化为行阶梯形矩阵【行阶梯形矩阵即越往下开头的 0 越多的矩阵】

【比如 $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 11 & 12 \\ 0 & 0 & 0 & 16 \end{bmatrix}$ 、 $\begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 5 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ 是行阶梯形矩阵】

【比如 $\begin{bmatrix} 8 & 8 & 4 & 8 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ 不是行阶梯形矩阵(阶梯太高了)】

消消乐的方式：①某行=本身-k·其他行【k可以是任何数】

②互换行与行

③某行·k【k可以是任何数】

消消乐之后：化成的行阶梯矩阵里有几行含有不是零的数，矩阵的秩就是几

练习1. 设矩阵 $A = \begin{bmatrix} 1 & -2 & -1 & 0 & 2 \\ -2 & 4 & 2 & 6 & -6 \\ 2 & -1 & 0 & 2 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{bmatrix}$ ，求 $r(A)$ 。

解： $A = \begin{bmatrix} 1 & -2 & -1 & 0 & 2 \\ -2 & 4 & 2 & 6 & -6 \\ 2 & -1 & 0 & 2 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{bmatrix} \xrightarrow{\substack{\text{行}_2+2\cdot\text{行}_1 \\ \text{行}_2-2\cdot\text{行}_1 \\ \text{行}_3-3\cdot\text{行}_1}} \begin{bmatrix} 1 & -2 & -1 & 0 & 2 \\ 0 & 0 & 0 & 6 & -2 \\ 0 & 3 & 2 & 2 & -1 \\ 0 & 9 & 6 & 3 & -2 \end{bmatrix} \xrightarrow{\text{行}_2\text{与行}_4\text{互换}} \begin{bmatrix} 1 & -2 & -1 & 0 & 2 \\ 0 & 9 & 6 & 3 & -2 \\ 0 & 3 & 2 & 2 & -1 \\ 0 & 0 & 0 & 6 & -2 \end{bmatrix}$

$\xrightarrow{\text{行}_3-\frac{1}{3}\cdot\text{行}_2} \begin{bmatrix} 1 & -2 & -1 & 0 & 2 \\ 0 & 9 & 6 & 3 & -2 \\ 0 & 0 & 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 0 & 6 & -2 \end{bmatrix} \xrightarrow{\text{行}_4-6\cdot\text{行}_3} \begin{bmatrix} 1 & -2 & -1 & 0 & 2 \\ 0 & 9 & 6 & 3 & -2 \\ 0 & 0 & 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore r(A)=3$

练习2. 设矩阵 $A = \begin{bmatrix} 1 & 2 & -2 & 3 \\ 4 & a & 3 & 12 \\ 3 & -1 & 1 & 9 \end{bmatrix}$ ，则 $a = \underline{\hspace{1cm}}$ 时 $r(A) < 3$ 。

解： $A = \begin{bmatrix} 1 & 2 & -2 & 3 \\ 4 & a & 3 & 12 \\ 3 & -1 & 1 & 9 \end{bmatrix} \xrightarrow{\substack{\text{行}_2-4\cdot\text{行}_1 \\ \text{行}_3-3\cdot\text{行}_1}} \begin{bmatrix} 1 & 2 & -2 & 3 \\ 0 & a-8 & 11 & 0 \\ 0 & -7 & 7 & 0 \end{bmatrix} \xrightarrow{\text{行}_2\text{与行}_3\text{互换}} \begin{bmatrix} 1 & 2 & -2 & 3 \\ 0 & -7 & 7 & 0 \\ 0 & a-8 & 11 & 0 \end{bmatrix} \xrightarrow{\text{行}_3+\frac{a-8}{7}\cdot\text{行}_2} \begin{bmatrix} 1 & 2 & -2 & 3 \\ 0 & -7 & 7 & 0 \\ 0 & 0 & a+3 & 0 \end{bmatrix}$

当 $a+3=0$ 时即 $a=-3$ 时， $r(A)=2$ ，此时满足 $r(A) < 3$

将待求向量组里的各个向量按列填成一个矩阵，这个矩阵的秩就是向量组的秩

练习1. 已知向量组 $\alpha_1=\begin{bmatrix}2\\0\\0\\2\end{bmatrix}, \alpha_2=\begin{bmatrix}0\\-1\\1\\0\end{bmatrix}, \alpha_3=\begin{bmatrix}0\\1\\-1\\0\end{bmatrix}, \alpha_4=\begin{bmatrix}-2\\3\\1\\-2\end{bmatrix}$ ，则该向量组的秩为__。

另一种问法. 已知向量组 $\alpha_1=\begin{bmatrix}2\\0\\0\\2\end{bmatrix}, \alpha_2=\begin{bmatrix}0\\-1\\1\\0\end{bmatrix}, \alpha_3=\begin{bmatrix}0\\1\\-1\\0\end{bmatrix}, \alpha_4=\begin{bmatrix}-2\\3\\1\\-2\end{bmatrix}$ ，则 $r(\alpha_1, \alpha_2, \alpha_3, \alpha_4)=$ __。

解：

$$\begin{bmatrix}2 & 0 & 0 & -2 \\ 0 & -1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 2 & 0 & 0 & -2\end{bmatrix} \xrightarrow{\text{行}_4-\text{行}_1} \begin{bmatrix}2 & 0 & 0 & -2 \\ 0 & -1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0\end{bmatrix} \xrightarrow{\text{行}_3+\text{行}_2} \begin{bmatrix}2 & 0 & 0 & -2 \\ 0 & -1 & 1 & 3 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0\end{bmatrix}$$

∴该向量组的秩为3
 $r(\alpha_1, \alpha_2, \alpha_3, \alpha_4)=3$

练习2. 已知向量组 $\alpha_1=(1,2,-1,1)^T, \alpha_2=(2,0,3,0)^T, \alpha_3=(0,-4,5,-2)^T, \alpha_4=(3,-2,7,-1)^T$ ，则该向量组的秩为__。

另一种问法. 已知向量组 $\alpha_1=(1,2,-1,1)^T, \alpha_2=(2,0,3,0)^T, \alpha_3=(0,-4,5,-2)^T, \alpha_4=(3,-2,7,-1)^T$ ，则 $r(\alpha_1, \alpha_2, \alpha_3, \alpha_4)=$ __。

解：

$$\begin{bmatrix}1 & 2 & 0 & 3 \\ 2 & 0 & -4 & -2 \\ -1 & 3 & 5 & 7 \\ 1 & 0 & -2 & -1\end{bmatrix} \xrightarrow{\begin{matrix} \text{行}_2-2\cdot\text{行}_1 \\ \text{行}_3+\text{行}_1 \\ \text{行}_4-\text{行}_1 \end{matrix}} \begin{bmatrix}1 & 2 & 0 & 3 \\ 0 & -4 & -4 & -8 \\ 0 & 5 & 5 & 10 \\ 0 & -2 & -2 & -4\end{bmatrix} \xrightarrow{\begin{matrix} \text{行}_3+\frac{5}{4}\cdot\text{行}_2 \\ \text{行}_4-\frac{1}{2}\cdot\text{行}_2 \end{matrix}} \begin{bmatrix}1 & 2 & 0 & 3 \\ 0 & -4 & -4 & -8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0\end{bmatrix}$$

∴该向量组的秩为2
 $r(\alpha_1, \alpha_2, \alpha_3, \alpha_4)=2$

知识点三：判断向量组的线性相关性 【什么时候是线性相关：有“大哥”可以由“小弟”组成】
【什么时候是线性无关：全是“小弟”，没有“大哥”】

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“小弟”数=向量组的秩

练习1. 设向量组 $\alpha_1=\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\alpha_2=\begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$, $\alpha_3=\begin{bmatrix} -1 \\ -3 \\ t \end{bmatrix}$, 当 $t=$ ___时 $\alpha_1, \alpha_2, \alpha_3$ 线性相关。

解：该向量组线性相关， $\therefore$ 有“大哥”（这是草稿
不写到卷面上）
总共有3个向量，其中有“大哥”
 $\therefore$ “小弟”数 <3
 $\therefore$ 向量组的秩 <3

$$\begin{bmatrix} 1 & -1 & -1 \\ 2 & -1 & -3 \\ 3 & 2 & t \end{bmatrix} \xrightarrow{\substack{\text{行}_2-2\cdot\text{行}_1 \\ \text{行}_3-3\cdot\text{行}_1}} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 5 & t+3 \end{bmatrix} \xrightarrow{\text{行}_3-5\cdot\text{行}_2} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & t+8 \end{bmatrix}$$

当 $t+8\neq0$ 时，秩=3

当 $t+8=0$ 时，秩=2

$\therefore$ 向量组的秩 <3

$\therefore t+8=0 \Rightarrow t=-8$

练习2. 已知向量组 $\alpha_1=(1,2,-1,1)^T$, $\alpha_2=(2,0,t,0)^T$, $\alpha_3=(0,-4,5,-2)^T$ 线性相关，则 $t=$ ___。

解：该向量组线性相关， $\therefore$ 有“大哥”（这是草稿
不写到卷面上）
总共有3个向量，其中有“大哥”
 $\therefore$ “小弟”数 <3
 $\therefore$ 向量组的秩 <3

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & -4 \\ -1 & t & 5 \\ 1 & 0 & -2 \end{bmatrix} \xrightarrow{\substack{\text{行}_2-2\cdot\text{行}_1 \\ \text{行}_3+\text{行}_1 \\ \text{行}_4-\text{行}_1}} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -4 & -4 \\ 0 & t+2 & 5 \\ 0 & -2 & -2 \end{bmatrix} \xrightarrow{\substack{\text{行}_3+\frac{t+2}{4}\cdot\text{行}_2 \\ \text{行}_4-\frac{1}{2}\cdot\text{行}_2}} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -4 & -4 \\ 0 & 0 & 3-t \\ 0 & 0 & 0 \end{bmatrix}$$

当 $3-t\neq0$ 时，秩=3

当 $3-t=0$ 时，秩=2

$\therefore$ 向量组的秩 <3

$\therefore 3-t=0 \Rightarrow t=3$

【若向量为“大哥”，则可以由其他向量线性表示】

“小弟”数=向量组的秩

练习1. 给定向量组 $\alpha_1=\begin{bmatrix}-2\\1\\0\\3\end{bmatrix}$, $\alpha_2=\begin{bmatrix}1\\-3\\2\\4\end{bmatrix}$, $\alpha_3=\begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$, $\alpha_4=\begin{bmatrix}0\\-1\\4\\9\end{bmatrix}$, 试判断 α_4 是否为 $\alpha_1, \alpha_2, \alpha_3$ 的线性组合。

解:
$$\begin{bmatrix}-2 & 1 & 3 \\ 1 & -3 & 0 \\ 0 & 2 & 2 \\ 3 & 4 & -1\end{bmatrix} \xrightarrow{\substack{\text{行}_1 \text{与行}_2 \text{互换} \\ \text{行}_2 \text{与行}_3 \text{互换}}} \begin{bmatrix}1 & -3 & 0 \\ 0 & 2 & 2 \\ -2 & 1 & 3 \\ 3 & 4 & -1\end{bmatrix} \xrightarrow{\substack{\text{行}_3+2\cdot\text{行}_1 \\ \text{行}_4-3\cdot\text{行}_1}} \begin{bmatrix}1 & -3 & 0 \\ 0 & 2 & 2 \\ 0 & -5 & 3 \\ 0 & 13 & -1\end{bmatrix}$$

$$\xrightarrow{\substack{\text{行}_3+\frac{5}{2}\cdot\text{行}_2 \\ \text{行}_4-\frac{13}{2}\cdot\text{行}_2}} \begin{bmatrix}1 & -3 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & -14\end{bmatrix} \xrightarrow{\text{行}_4+\frac{7}{4}\cdot\text{行}_3} \begin{bmatrix}1 & -3 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 0\end{bmatrix} \Rightarrow \text{向量组的秩为3} \Rightarrow \alpha_1, \alpha_2, \alpha_3 \text{里有3个小弟}$$

$$\begin{bmatrix}-2 & 1 & 3 & 0 \\ 1 & -3 & 0 & -1 \\ 0 & 2 & 2 & 4 \\ 3 & 4 & -1 & 9\end{bmatrix} \xrightarrow{\substack{\text{行}_1 \text{与行}_2 \text{互换} \\ \text{行}_2 \text{与行}_3 \text{互换}}} \begin{bmatrix}1 & -3 & 0 & -1 \\ 0 & 2 & 2 & 4 \\ -2 & 1 & 3 & 0 \\ 3 & 4 & -1 & 9\end{bmatrix} \xrightarrow{\substack{\text{行}_3+2\cdot\text{行}_1 \\ \text{行}_4-3\cdot\text{行}_1}} \begin{bmatrix}1 & -3 & 0 & -1 \\ 0 & 2 & 2 & 4 \\ 0 & -5 & 3 & -2 \\ 0 & 13 & -1 & 12\end{bmatrix}$$

$$\xrightarrow{\substack{\text{行}_3+\frac{5}{2}\cdot\text{行}_2 \\ \text{行}_4-\frac{13}{2}\cdot\text{行}_2}} \begin{bmatrix}1 & -3 & 0 & -1 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & 8 & 8 \\ 0 & 0 & -14 & -14\end{bmatrix} \xrightarrow{\text{行}_4+\frac{7}{4}\cdot\text{行}_3} \begin{bmatrix}1 & -3 & 0 & -1 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & 8 & 8 \\ 0 & 0 & 0 & 0\end{bmatrix} \Rightarrow \text{秩为3} \Rightarrow \alpha_1, \alpha_2, \alpha_3, \alpha_4 \text{里有3个小弟}$$

$\because \alpha_1, \alpha_2, \alpha_3$ 里有3个小弟, $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 里有3个小弟 $\Rightarrow \alpha_4$ 是大哥

$\therefore \alpha_4$ 能被 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

综上, 秩相等 $\Rightarrow \alpha_4$ 是 $\alpha_1, \alpha_2, \alpha_3$ 的线性组合

练习2. 已知向量 $\alpha_1=\begin{bmatrix}1\\2\\1\end{bmatrix}$, $\alpha_2=\begin{bmatrix}1\\3\\4\end{bmatrix}$, $\beta=\begin{bmatrix}1\\1\\a\end{bmatrix}$, 当a=___时 β 可被 α_1, α_2 线性表示。

解: β 可被 α_1, α_2 线性表示 $\Rightarrow \beta$ 是大哥

$$\begin{bmatrix}1 & 1 \\ 2 & 3 \\ 1 & 4\end{bmatrix} \xrightarrow{\substack{\text{行}_2-2\cdot\text{行}_1 \\ \text{行}_3-\text{行}_1}} \begin{bmatrix}1 & 1 \\ 0 & 1 \\ 0 & 3\end{bmatrix} \xrightarrow{\text{行}_3-3\cdot\text{行}_2} \begin{bmatrix}1 & 1 \\ 0 & 1 \\ 0 & 0\end{bmatrix} \Rightarrow \text{秩为2} \Rightarrow \alpha_1, \alpha_2 \text{里有2个小弟}$$

$$\begin{bmatrix}1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 4 & a\end{bmatrix} \xrightarrow{\substack{\text{行}_2-2\cdot\text{行}_1 \\ \text{行}_3-\text{行}_1}} \begin{bmatrix}1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 3 & a-1\end{bmatrix} \xrightarrow{\text{行}_3-3\cdot\text{行}_2} \begin{bmatrix}1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & a+2\end{bmatrix}$$

当 $a+2 \neq 0$ 时, 秩=3 $\Rightarrow \alpha_1, \alpha_2, \beta$ 里有3个小弟, $\because \alpha_1, \alpha_2$ 里有2个小弟 $\therefore \beta$ 是小弟 (不符合要求)

当 $a+2 = 0$ 时, 秩=2 $\Rightarrow \alpha_1, \alpha_2, \beta$ 里有2个小弟, $\because \alpha_1, \alpha_2$ 里有2个小弟 $\therefore \beta$ 是大哥 (符合要求)

$\therefore a+2=0 \Rightarrow a=-2$

求完秩后，每行第一个非零数所在列对应的向量就是一个极(最)大无关组

练习1. 设有向量组 $\alpha_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 7 \\ 0 \\ 14 \\ 3 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 5 \\ 1 \\ 6 \\ 2 \end{bmatrix}$ 。

(1)求该向量组的秩；(2)求该向量组的一个最大无关组。

解：

$$\begin{bmatrix} 1 & 7 & 2 & 5 \\ 3 & 0 & -1 & 1 \\ 2 & 14 & 0 & 6 \\ 0 & 3 & 1 & 2 \end{bmatrix} \xrightarrow{\substack{\text{行}_2-3\cdot\text{行}_1 \\ \text{行}_3-2\cdot\text{行}_1}} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & -21 & -7 & -14 \\ 0 & 0 & -4 & -4 \\ 0 & 3 & 1 & 2 \end{bmatrix} \xrightarrow{\text{行}_4+\frac{1}{7}\cdot\text{行}_2} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & -21 & -7 & -14 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{matrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \\ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 7 \\ -21 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 2 \\ -7 \\ -4 \\ 0 \end{bmatrix} & \begin{bmatrix} 5 \\ -14 \\ -4 \\ 0 \end{bmatrix} \end{matrix}$$

∴该向量组的秩为3，一个最大无关组是 $\alpha_1, \alpha_2, \alpha_3$

练习2. 已知向量组 $\alpha_1=(1,0,3,1)^T, \alpha_2=(2,1,7,2)^T, \alpha_3=(-1,2,0,-1)^T, \alpha_4=(1,4,5,a)^T$ ，请问a为何值时该向量组线性相关？
此时求该向量组的一个极大无关组。

解：该向量组线性相关，∴有“大哥”（这是草稿，不写到卷面上）
总共有4个向量，其中有“大哥”
∴“小弟”数<4
∴向量组的秩<4

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 3 & 7 & 0 & 5 \\ 1 & 2 & -1 & a \end{bmatrix} \xrightarrow{\substack{\text{行}_3-3\cdot\text{行}_1 \\ \text{行}_4-\text{行}_1}} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & a-1 \end{bmatrix} \xrightarrow{\text{行}_3-\text{行}_2} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & a-1 \end{bmatrix}$$

当 $a-1 \neq 0$ 时，秩=4（不符合要求）

当 $a-1=0$ 时，秩=3（符合要求）

∴ $a-1=0 \Rightarrow a=1$

$$\begin{matrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \\ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 1 \\ 4 \\ -2 \\ 0 \end{bmatrix} \end{matrix}$$

a=1时行阶梯矩阵为

∴一个极大线性无关组是 $\alpha_1, \alpha_2, \alpha_3$

(1)求秩

(2)判断是否有反常规情况

若有 $\Rightarrow$ 无解

若没有，则有效式子 = 秩、且 $\begin{cases} \text{有效式子} = \text{未知数个数} \Rightarrow \text{有一组解} \\ \text{有效式子} < \text{未知数个数} \Rightarrow \text{有无穷多组解} \end{cases}$

【某行虚线左侧全是0，虚线右侧不是0】

练习1. 已知方程组 $\begin{cases} x_1 + 2x_2 + x_3 = 1 \\ 2x_1 + 3x_2 + (a + 2)x_3 = 3 \\ x_1 + ax_2 - 2x_3 = 0 \end{cases}$ 无解，则a=___。

解： $\begin{cases} x_1 + 2x_2 + x_3 = 1 \\ 2x_1 + 3x_2 + (a + 2)x_3 = 3 \\ x_1 + ax_2 - 2x_3 = 0 \end{cases} \Rightarrow \begin{cases} 1x_1 + 2x_2 + 1x_3 = 1 \\ 2x_1 + 3x_2 + (a + 2)x_3 = 3 \\ 1x_1 + ax_2 - 2x_3 = 0 \end{cases}$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 3 & a+2 & 3 \\ 1 & a & -2 & 0 \end{array} \right] \xrightarrow[\text{行}_3 - \text{行}_1]{\text{行}_2 - 2 \cdot \text{行}_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & a & 1 \\ 0 & a-2 & -3 & -1 \end{array} \right] \xrightarrow{\text{行}_3 + (a-2) \cdot \text{行}_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & a & 1 \\ 0 & 0 & a^2 - 2a - 3 & a - 3 \end{array} \right]$$

∴ 该方程组无解

∴ 存在反常规情况

∴ $a^2 - 2a - 3 = 0$ 且 $a - 3 \neq 0 \Rightarrow a = -1$

练习2. 已知线性方程组 $\begin{cases} x_1 + 3x_2 + x_3 = 0 \\ 3x_1 + 2x_2 + 3x_3 = -7 \\ -x_1 + 4x_2 + mx_3 = k \end{cases}$ ，问m、k各取何值时，方程组无解？有唯一解？有无穷多组解？

解： $\begin{cases} x_1 + 3x_2 + x_3 = 0 \\ 3x_1 + 2x_2 + 3x_3 = -7 \\ -x_1 + 4x_2 + mx_3 = k \end{cases} \Rightarrow \begin{cases} 1x_1 + 3x_2 + 1x_3 = 0 \\ 3x_1 + 2x_2 + 3x_3 = -7 \\ -1x_1 + 4x_2 + mx_3 = k \end{cases}$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 3 & 2 & 3 & -7 \\ -1 & 4 & m & k \end{array} \right] \xrightarrow[\text{行}_3 + \text{行}_1]{\text{行}_2 - 3 \cdot \text{行}_1} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 7 & m+1 & k \end{array} \right] \xrightarrow{\text{行}_3 + \text{行}_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & m+1 & k-7 \end{array} \right]$$

当 $m+1 \neq 0$ 时，即当 $m \neq -1$ 时，行阶梯矩阵为 $\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & \text{不为0} & k-7 \end{array} \right]$ ，矩阵的秩为3，有效式子=3=未知数个数，有一组解

当 $m+1=0$ 且 $k-7=0$ 时，即当 $m=-1$ 且 $k=7$ 时，行阶梯矩阵为 $\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{array} \right]$ ，矩阵的秩为2，有效式子=2<未知数个数，

有无穷多组解

当 $m+1=0$ 且 $k-7 \neq 0$ 时，即当 $m=-1$ 且 $k \neq 7$ 时，行阶梯矩阵为 $\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & 0 & \text{不为0} \end{array} \right]$ ，有反常规情况，方程组无解

处理矩阵的方式：①某行=本身-k·其他行【k可以是任何数】

②互换行与行

③某行·k【k可以是任何数】

处理矩阵的顺序：(1)得到行阶梯形矩阵

(2)使每行首个非零数为1

(3)使每行首个非零数所在的列里其余数都是0

练习1. 设有向量组 $\alpha_1=\begin{bmatrix} 1 \\ 3 \\ 2 \\ 0 \end{bmatrix}, \alpha_2=\begin{bmatrix} 7 \\ 0 \\ 14 \\ 3 \end{bmatrix}, \alpha_3=\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \alpha_4=\begin{bmatrix} 5 \\ 1 \\ 6 \\ 2 \end{bmatrix}$ ，试将向量组化为行最简型矩阵。

解：

$$\begin{bmatrix} 1 & 7 & 2 & 5 \\ 3 & 0 & -1 & 1 \\ 2 & 14 & 0 & 6 \\ 0 & 3 & 1 & 2 \end{bmatrix} \xrightarrow{\substack{\text{行}_2-3\cdot\text{行}_1 \\ \text{行}_3-2\cdot\text{行}_1}} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & -21 & -7 & -14 \\ 0 & 0 & -4 & -4 \\ 0 & 3 & 1 & 2 \end{bmatrix} \xrightarrow{\text{行}_4+\frac{1}{7}\cdot\text{行}_2} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & -21 & -7 & -14 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$
$$\xrightarrow{\substack{\text{行}_2\cdot(-\frac{1}{21}) \\ \text{行}_3\cdot(-\frac{1}{4})}} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{行}_1-7\cdot\text{行}_2} \begin{bmatrix} 1 & 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{\text{行}_1+\frac{1}{3}\cdot\text{行}_3 \\ \text{行}_2-\frac{1}{3}\cdot\text{行}_3}} \begin{bmatrix} 1 & 0 & 0 & \frac{2}{3} \\ 0 & 1 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

练习2. 已知向量组 $\alpha_1=(1,0,3,1)^T, \alpha_2=(2,1,7,2)^T, \alpha_3=(-1,2,0,-1)^T, \alpha_4=(1,4,5,1)^T$ ，试将向量组化为行最简型矩阵。

解：

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 3 & 7 & 0 & 5 \\ 1 & 2 & -1 & 1 \end{bmatrix} \xrightarrow{\substack{\text{行}_3-3\cdot\text{行}_1 \\ \text{行}_4-\text{行}_1}} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{行}_3-\text{行}_2} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{行}_1-2\cdot\text{行}_2} \begin{bmatrix} 1 & 0 & -5 & -7 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{\text{行}_1+5\cdot\text{行}_3 \\ \text{行}_2-2\cdot\text{行}_3}} \begin{bmatrix} 1 & 0 & 0 & -17 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(1)将向量组构成一个矩阵，再化成行最简形矩阵

(2)找出极大无关组：每行第一个非零数所在列对应的向量

(3)其余向量 = 该向量对应列的数₁ · 极大无关组里第一个向量
+ 数₂ · 极大无关组里第二个向量
+ ...

练习1. 设有向量组 $\alpha_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 7 \\ 0 \\ 14 \\ 3 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 5 \\ 1 \\ 6 \\ 2 \end{bmatrix}$ 。

(1)求该向量组的秩；(2)求该向量组的一个极大无关组，并把其余向量分别用求得的最大无关组表示出来

解：
$$\begin{bmatrix} 1 & 7 & 2 & 5 \\ 3 & 0 & -1 & 1 \\ 2 & 14 & 0 & 6 \\ 0 & 3 & 1 & 2 \end{bmatrix} \xrightarrow{\substack{\text{行}_2 - 3 \cdot \text{行}_1 \\ \text{行}_3 - 2 \cdot \text{行}_1}} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & -21 & -7 & -14 \\ 0 & 0 & -4 & -4 \\ 0 & 3 & 1 & 2 \end{bmatrix} \xrightarrow{\text{行}_4 + \frac{1}{7} \cdot \text{行}_2} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & -21 & -7 & -14 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4$

$$\xrightarrow{\substack{\text{行}_2 \cdot (-\frac{1}{21}) \\ \text{行}_3 \cdot (-\frac{1}{4})}} \begin{bmatrix} 1 & 7 & 2 & 5 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{行}_1 - 7 \cdot \text{行}_2} \begin{bmatrix} 1 & 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{\text{行}_1 + \frac{1}{3} \cdot \text{行}_3 \\ \text{行}_2 - \frac{1}{3} \cdot \text{行}_3}} \begin{bmatrix} 1 & 0 & 0 & \frac{2}{3} \\ 0 & 1 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

∴该向量组的秩为3，一个极大无关组是 $\alpha_1, \alpha_2, \alpha_3$

$$\alpha_4 = \frac{2}{3}\alpha_1 + \frac{1}{3}\alpha_2 + 1\alpha_3 + 0 = \frac{2}{3}\alpha_1 + \frac{1}{3}\alpha_2 + \alpha_3$$

练习2. 已知向量组 $\alpha_1 = (1, 0, 3, 1)^T, \alpha_2 = (2, 1, 7, 2)^T, \alpha_3 = (-1, 2, 0, -1)^T, \alpha_4 = (1, 4, 5, a)^T$ ，请问a为何值时该向量组线性相关？

此时求该向量组的一个极大无关组，并将其余向量用此极大无关组表出

解：该向量组线性相关，∴有“大哥”（这是草稿，不写到卷面上）

总共有4个向量，其中有“大哥”

∴“小弟”数 < 4

∴向量组的秩 < 4

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 3 & 7 & 0 & 5 \\ 1 & 2 & -1 & a \end{bmatrix} \xrightarrow{\substack{\text{行}_3 - 3 \cdot \text{行}_1 \\ \text{行}_4 - \text{行}_1}} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & a-1 \end{bmatrix} \xrightarrow{\text{行}_3 - \text{行}_2} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & a-1 \end{bmatrix}$$

当 $a-1 \neq 0$ 时，秩=4（不符合要求）

当 $a-1=0$ 时，秩=3（符合要求）

∴ $a-1=0 \Rightarrow a=1$

$\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4$

$$a=1 \text{ 时矩阵} = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{行}_1 - 2 \cdot \text{行}_2} \begin{bmatrix} 1 & 0 & -5 & -7 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{\text{行}_1 + 5 \cdot \text{行}_3 \\ \text{行}_2 - 2 \cdot \text{行}_3}} \begin{bmatrix} 1 & 0 & 0 & -17 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

∴一个极大线性无关组是 $\alpha_1, \alpha_2, \alpha_3$

$$\alpha_4 = -17\alpha_1 + 8\alpha_2 - 2\alpha_3 + 0 = -17\alpha_1 + 8\alpha_2 - 2\alpha_3$$

(1)写出方程组的矩阵，将矩阵化成行最简形矩阵

(2)将行最简形矩阵还原成大括号的形式

(3)化简大括号，然后将各个式子的等号左边只保留第一项，其余内容都移到等号右边

(4)补上缺的未知数式子，将各个式子等号右边写成规范的格式，调整各个式子的顺序

(5)写出通解

练习1. 求线性方程组
$$\begin{cases} 2x_1 - x_2 + 4x_3 - 3x_4 = -4 \\ x_1 + x_3 - x_4 = -3 \\ 3x_1 + x_2 + x_3 = 1 \\ 7x_1 + 7x_3 - 3x_4 = 3 \end{cases}$$
 的通解

解：
$$\left[\begin{array}{cccc|c} 2 & -1 & 4 & -3 & -4 \\ 1 & 0 & 1 & -1 & -3 \\ 3 & 1 & 1 & 0 & 1 \\ 7 & 0 & 7 & -3 & 3 \end{array} \right] \xrightarrow{\text{行}_1 \text{与行}_2 \text{交换}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -3 \\ 2 & -1 & 4 & -3 & -4 \\ 3 & 1 & 1 & 0 & 1 \\ 7 & 0 & 7 & -3 & 3 \end{array} \right]$$

$$\xrightarrow{\substack{\text{行}_2 - 2 \cdot \text{行}_1 \\ \text{行}_3 - 3 \cdot \text{行}_1 \\ \text{行}_4 - 7 \cdot \text{行}_1}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -3 \\ 0 & -1 & 2 & -1 & 2 \\ 0 & 1 & -2 & 3 & 10 \\ 0 & 0 & 0 & 4 & 24 \end{array} \right] \xrightarrow{\text{行}_3 + \text{行}_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -3 \\ 0 & -1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 2 & 12 \\ 0 & 0 & 0 & 4 & 24 \end{array} \right] \xrightarrow{\text{行}_4 - 2 \cdot \text{行}_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -3 \\ 0 & -1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 2 & 12 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{\substack{\text{行}_2 \cdot (-1) \\ \text{行}_3 \cdot \frac{1}{2}}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -3 \\ 0 & 1 & -2 & 1 & -2 \\ 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\substack{\text{行}_1 + \text{行}_3 \\ \text{行}_2 - \text{行}_3}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & -2 & 0 & -8 \\ 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \begin{cases} 1x_1 + 0x_2 + 1x_3 + 0x_4 = 3 \\ 0x_1 + 1x_2 - 2x_3 + 0x_4 = -8 \\ 0x_1 + 0x_2 + 0x_3 + 1x_4 = 6 \\ 0x_1 + 0x_2 + 0x_3 + 0x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 + x_3 = 3 \\ x_2 - 2x_3 = -8 \\ x_4 = 6 \end{cases} \Rightarrow \begin{cases} x_1 = 3 - x_3 \\ x_2 = -8 + 2x_3 \\ x_4 = 6 \end{cases} \Rightarrow \begin{cases} x_1 = 3 + (-1) \cdot x_3 \\ x_2 = -8 + 2 \cdot x_3 \\ x_3 = x_3 \\ x_4 = 6 + 0 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ -8 \\ 0 \\ 6 \end{bmatrix} + k \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \text{其中} k \text{为任意常数}$$

练习2. 已知线性方程组
$$\begin{cases} x_1 + 3x_2 + x_3 = 0 \\ 3x_1 + 2x_2 + 3x_3 = -7 \\ -x_1 + 4x_2 + mx_3 = k \end{cases}$$
，问m、k各取何值时，方程组无解？有唯一解？有无穷多组解？

并求有无穷多组解时的通解。

解：
$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 3 & 2 & 3 & -7 \\ -1 & 4 & m & k \end{array} \right] \xrightarrow{\substack{\text{行}_2 - 3 \cdot \text{行}_1 \\ \text{行}_3 + \text{行}_1}} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 7 & m+1 & k \end{array} \right] \xrightarrow{\text{行}_3 + \text{行}_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & m+1 & k-7 \end{array} \right]$$

当m+1≠0时，即当m≠-1时，行阶梯矩阵为
$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & \text{不为0} & k-7 \end{array} \right]$$
，矩阵的秩为3，有效式子=3=未知数个数，有一组解

当m+1=0且k-7=0时，即当m=-1且k=7时，行阶梯矩阵为
$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{array} \right]$$
，矩阵的秩为2，有效式子=2<未知数个数，

有无穷多组解

当m+1=0且k-7≠0时，即当m=-1且k≠7时，行阶梯矩阵为
$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & 0 & \text{不为0} \end{array} \right]$$
，有反常规情况，方程组无解

当m=-1且k=7时，矩阵为
$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\text{行}_2 \cdot (-\frac{1}{7})} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\text{行}_1 - 3 \cdot \text{行}_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow \begin{cases} 1x_1 + 0x_2 + 1x_3 = -3 \\ 0x_1 + 1x_2 + 0x_3 = 1 \\ 0x_1 + 0x_2 + 0x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 + x_3 = -3 \\ x_2 = 1 \end{cases} \Rightarrow \begin{cases} x_1 = -3 - x_3 \\ x_2 = 1 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = -3 + (-1) \cdot x_3 \\ x_2 = 1 + 0 \cdot x_3 \\ x_3 = 0 + 1 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} + k \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \text{其中} k \text{为任意常数}$$

(1)待求矩阵填到大矩阵的左半部分，画上虚线，再在虚线右侧填 **1、0**（对角线为**1**，其余为**0**）

(2)化大矩阵为行最简形矩阵

(3)结果里**虚线右侧的矩阵**就是逆矩阵

练习1. 设矩阵 $A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ ，求 A 的逆矩阵

另一种问法. 设矩阵 $A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ ，求 A^{-1}

$$\begin{aligned} \text{解: } \left[\begin{array}{ccc|ccc} 2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & 1 & 0 & 0 & 1 \end{array} \right] & \xrightarrow{\substack{\text{行}_1 \text{与行}_2 \text{互换} \\ \text{行}_2 \text{与行}_3 \text{互换}}} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & 1 & 0 & 0 & 1 \\ 2 & 2 & 3 & 1 & 0 & 0 \end{array} \right] \\ & \xrightarrow{\substack{\text{行}_2 + \text{行}_1 \\ \text{行}_3 - 2 \cdot \text{行}_1}} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 4 & 3 & 1 & -2 & 0 \end{array} \right] \xrightarrow{\text{行}_3 - 4 \cdot \text{行}_2} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 & -6 & -4 \end{array} \right] \\ & \xrightarrow{\text{行}_3 \cdot (-1)} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 & 6 & 4 \end{array} \right] \xrightarrow{\text{行}_1 + \text{行}_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 & 6 & 4 \end{array} \right] \xrightarrow{\substack{\text{行}_1 - \text{行}_3 \\ \text{行}_2 - \text{行}_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -4 & -3 \\ 0 & 1 & 0 & 1 & -5 & -3 \\ 0 & 0 & 1 & -1 & 6 & 4 \end{array} \right] \\ \therefore A^{-1} &= \begin{bmatrix} 1 & -4 & -3 \\ 1 & -5 & -3 \\ -1 & 6 & 4 \end{bmatrix} \end{aligned}$$

练习2. 设 $A = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 3 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ ，求 A 的逆矩阵

另一种问法. 设 $A = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 3 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ ，求 A^{-1}

$$\begin{aligned} \text{解: } \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 0 \\ 1 & 3 & 4 & 0 & 0 & 1 \end{array} \right] & \xrightarrow{\substack{\text{行}_2 - 2 \cdot \text{行}_1 \\ \text{行}_3 - \text{行}_1}} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 0 & -3 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right] \\ & \xrightarrow{\text{行}_2 \cdot (-\frac{1}{3})} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right] \xrightarrow{\text{行}_1 - 3 \cdot \text{行}_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right] \xrightarrow{\text{行}_2 - \text{行}_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & \frac{5}{3} & -\frac{1}{3} & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right] \\ \therefore A^{-1} &= \begin{bmatrix} -1 & 1 & 0 \\ \frac{5}{3} & -\frac{1}{3} & -1 \\ -1 & 0 & 1 \end{bmatrix} \end{aligned}$$

二阶行列式： $\begin{vmatrix} A & B \\ C & D \end{vmatrix} = AD - BC$ 【比如 $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \times 4 - 2 \times 3 = -2$ 】

【比如 $\begin{vmatrix} 5 & 7 \\ 6 & 9 \end{vmatrix} = 5 \times 9 - 7 \times 6 = 3$ 】

多阶行列式：通过方式①、②、③将其化成 $\begin{vmatrix} ? & ? & ? \\ 0 & ? & ? \\ 0 & 0 & ? \end{vmatrix}$ 的格式后，对角线上各个元素的乘积就是行列式的值

方式①某行(列)的公因子可以提到| |外

方式②某行(列)=本身-k·其他行(列)

方式③互换行(列)与行(列)，| |前乘-1

练习1. 计算 $D = \begin{vmatrix} 2 & 1 & 4 & 1 \\ 3 & -1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 6 & 0 & 7 & 2 \end{vmatrix}$

解：D 行₁与行₃互换 $\begin{vmatrix} 1 & 2 & 3 & 2 \\ 3 & -1 & 2 & 1 \\ 2 & 1 & 4 & 1 \\ 6 & 0 & 7 & 2 \end{vmatrix}$ 【注：此时第一行就变得比较简单了】

行₂-3·行₁
行₃-2·行₁
行₄-6·行₁ $\begin{vmatrix} 1 & 2 & 3 & 2 \\ 0 & -7 & -7 & -5 \\ 0 & -3 & -2 & -3 \\ 0 & -12 & -11 & -10 \end{vmatrix}$ 行₃ - $\frac{3}{7}$ ·行₂
行₄ - $\frac{12}{7}$ ·行₂ $\begin{vmatrix} 1 & 2 & 3 & 2 \\ 0 & -7 & -7 & -5 \\ 0 & 0 & 1 & -\frac{6}{7} \\ 0 & 0 & 1 & -\frac{10}{7} \end{vmatrix}$ 行₄-行₃ $\begin{vmatrix} 1 & 2 & 3 & 2 \\ 0 & -7 & -7 & -5 \\ 0 & 0 & 1 & -\frac{6}{7} \\ 0 & 0 & 0 & -\frac{4}{7} \end{vmatrix}$

$= -[1 \times (-7) \times 1 \times (-\frac{4}{7})] = -4$

练习2. 计算行列式 $\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 2 & 1 & \cdots & 1 & 1 \\ 1 & 1 & 3 & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & n-1 & 1 \\ 1 & 1 & 1 & \cdots & 1 & n \end{vmatrix}$

解： $\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 2 & 1 & \cdots & 1 & 1 \\ 1 & 1 & 3 & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & n-1 & 1 \\ 1 & 1 & 1 & \cdots & 1 & n \end{vmatrix}$ 行₂-行₁
行₃-行₁
⋮
行_n-行₁ $\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & n-2 & 0 \\ 0 & 0 & 0 & \cdots & 0 & n-1 \end{vmatrix} = 1 \times 1 \times 2 \times \cdots (n-2) \cdot (n-1) = (n-1)!$

知识点12：判断一个矩阵可不可逆 / 是不是可逆矩阵

B站：猴博士爱讲课
有相关的免费视频

求|矩阵里的元素|，若结果 = 0，则 不可逆 / 不是可逆矩阵
若结果 ≠ 0，则 可逆 / 是可逆矩阵

练习1. 设 $A=\begin{bmatrix}1 & 3 & 3 \\ 2 & 3 & 3 \\ 1 & 3 & 4\end{bmatrix}$ ，证明矩阵 A 可逆并求 A^{-1}

解：∵ $\begin{vmatrix}1 & 3 & 3 \\ 2 & 3 & 3 \\ 1 & 3 & 4\end{vmatrix} \xrightarrow[\text{行}_3-\text{行}_1]{\text{行}_2-2\cdot\text{行}_1} \begin{vmatrix}1 & 3 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & 1\end{vmatrix} = 1 \times (-3) \times 1 = -3 \neq 0$ ，∴ 矩阵 A 可逆

$$\begin{bmatrix}1 & 3 & 3 & | & 1 & 0 & 0 \\ 2 & 3 & 3 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1\end{bmatrix} \xrightarrow[\text{行}_3-\text{行}_1]{\text{行}_2-2\cdot\text{行}_1} \begin{bmatrix}1 & 3 & 3 & | & 1 & 0 & 0 \\ 0 & -3 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1\end{bmatrix}$$
$$\xrightarrow{\text{行}_2 \cdot (-\frac{1}{3})} \begin{bmatrix}1 & 3 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1\end{bmatrix} \xrightarrow{\text{行}_1-3\cdot\text{行}_2} \begin{bmatrix}1 & 0 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1\end{bmatrix} \xrightarrow{\text{行}_2-\text{行}_3} \begin{bmatrix}1 & 0 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & 0 & | & \frac{5}{3} & -\frac{1}{3} & -1 \\ 0 & 0 & 1 & | & -1 & 0 & 1\end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ \frac{5}{3} & -\frac{1}{3} & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

知识点13：求矩阵的特征值

B站：猴博士爱讲课

有相关的免费视频

(1)把矩阵里的元素拿出来，**对角线**减去 λ ，构成行列式

(2)求行列式的值

(3)令行列式的值=0 $\Rightarrow \begin{cases} \lambda_1 = ? \\ \lambda_2 = ? \\ \lambda_3 = ? \end{cases}$ 【 λ 个数等于矩阵的行数或列数 (要关注行列式结果里的指数)】

练习1. 设矩阵 $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ，求矩阵 A 的特征值

$$\begin{aligned} \text{解: } A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} &\Rightarrow \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{vmatrix} 1 & 2-\lambda & 0 \\ 2-\lambda & 1 & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_2 - (2-\lambda) \cdot \text{行}_1} \begin{vmatrix} 1 & 2-\lambda & 0 \\ 0 & (3-\lambda)(\lambda-1) & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \\ &= -[1 \cdot (3-\lambda)(\lambda-1) \cdot (3-\lambda)] \\ &= -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1) \\ \text{令 } -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1) &= 0 \\ \Rightarrow \lambda_1 = 3, \lambda_2 = 3, \lambda_3 = 1 &\Rightarrow \text{特征值: } \lambda_1 = \lambda_2 = 3, \lambda_3 = 1 \end{aligned}$$

练习2. 设矩阵 $A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$ ，求矩阵 A 的特征值

$$\begin{aligned} \text{解: } A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} &\Rightarrow \begin{vmatrix} 3-\lambda & -1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & -1 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 3-\lambda & -1 & -1 \\ -1 & -1 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_2 + (3-\lambda) \cdot \text{行}_1} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \\ 0 & \lambda-4 & 4-\lambda \end{vmatrix} \\ &\xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \end{vmatrix} \xrightarrow{\text{行}_3 - (\lambda-2) \cdot \text{行}_2} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & 0 & (\lambda-4)(\lambda-1) \end{vmatrix} \\ &= -1 \cdot (\lambda-4) \cdot (\lambda-4) \cdot (\lambda-1) \\ &= -(\lambda-1)(\lambda-4)(\lambda-4) \\ \text{令 } -(\lambda-1)(\lambda-4)(\lambda-4) &= 0 \\ \Rightarrow \lambda_1 = 1, \lambda_2 = 4, \lambda_3 = 4 &\Rightarrow \text{特征值: } \lambda_1 = 1, \lambda_2 = \lambda_3 = 4 \end{aligned}$$

(1)求特征值

(2)对每一个不同的特征值都进行下面的操作：

- 将矩阵**对角线的每个元素**均减 λ ，再在右侧画虚线、加上一列0，构成新矩阵
- 将新矩阵化为行最简形矩阵
- 将行最简形矩阵变为方程，求通解
- 通解即为该 λ 对应的特征向量 **【k不能全取0】**

练习1. 设矩阵 $A=\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ，求矩阵 A 的特征值与特征向量

$$\begin{aligned} \text{解： } A &= \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{bmatrix} 1 & 2-\lambda & 0 \\ 2-\lambda & 1 & 0 \\ 0 & 0 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_2 - (2-\lambda) \cdot \text{行}_1} \begin{bmatrix} 1 & 2-\lambda & 0 \\ 0 & (3-\lambda)(\lambda-1) & 0 \\ 0 & 0 & 3-\lambda \end{bmatrix} \\ &= -[1 \cdot (3-\lambda)(\lambda-1) \cdot (3-\lambda)] \\ &= -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1) \\ &\text{令 } -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1) = 0 \\ &\Rightarrow \lambda_1=3, \lambda_2=3, \lambda_3=1 \Rightarrow \text{特征值： } \lambda_1=\lambda_2=3, \lambda_3=1 \end{aligned}$$

$$\begin{aligned} \text{对于 } \lambda_1=\lambda_2=3: & \begin{bmatrix} 2-3 & 1 & 0 & | & 0 \\ 1 & 2-3 & 0 & | & 0 \\ 0 & 0 & 3-3 & | & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 + x_2 + 0 \cdot x_3 = 0 \\ x_1 - x_2 + 0 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 0 \end{cases} \\ & \begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 + \text{行}_1} \begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_1 \cdot (-1)} \begin{bmatrix} 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ & \Rightarrow \{1x_1 - 1x_2 + 0x_3 = 0\} \Rightarrow \{x_1 = x_2\} \Rightarrow \begin{cases} x_1 = x_2 \\ x_2 = x_2 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_2 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_3 = 0 \cdot x_2 + 1 \cdot x_3 \end{cases} \\ & \Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \text{ 其中 } k_1, k_2 \text{ 为任意常数} \\ & \Rightarrow \text{特征向量为 } k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ (} k_1, k_2 \text{ 不同时为零)} \end{aligned}$$

$$\begin{aligned} \text{对于 } \lambda_3=1: & \begin{bmatrix} 2-1 & 1 & 0 & | & 0 \\ 1 & 2-1 & 0 & | & 0 \\ 0 & 0 & 3-1 & | & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 + x_2 + 0 \cdot x_3 = 0 \\ x_1 + x_2 + 0 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 2x_3 = 0 \end{cases} \\ & \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 - \text{行}_1} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 \cdot \frac{1}{2}} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ & \Rightarrow \begin{cases} 1x_1 + 1x_2 + 0x_3 = 0 \\ 0x_1 + 0x_2 + 1x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -x_2 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \cdot x_2 \\ x_2 = 1 \cdot x_2 \\ x_3 = 0 \cdot x_2 \end{cases} \\ & \Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \text{ 其中 } k_3 \text{ 为任意常数} \\ & \Rightarrow \text{特征向量为 } k_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \text{ (} k_3 \neq 0 \text{)} \end{aligned}$$

(1)求特征值

(2)对每一个不同的特征值都进行下面的操作：

- 将矩阵**对角线的每个元素**均减 λ ，再在右侧画虚线、加上一列0，构成新矩阵
- 将新矩阵化为行最简形矩阵
- 将行最简形矩阵变为方程，求通解
- 通解即为该 λ 对应的特征向量 **【k不能全取0】**

练习2. 设矩阵 $A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$ ，求矩阵 A 的特征值与特征向量

$$\begin{aligned} \text{解： } A &= \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 3-\lambda & -1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & -1 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 3-\lambda & -1 & -1 \\ -1 & -1 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_2 + (3-\lambda) \cdot \text{行}_1} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \\ 0 & \lambda-4 & 4-\lambda \end{bmatrix} \\ &\xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \end{bmatrix} \xrightarrow{\text{行}_3 - (\lambda-2) \cdot \text{行}_2} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & 0 & (\lambda-4)(\lambda-1) \end{bmatrix} \\ &= -1 \cdot (\lambda-4) \cdot (\lambda-4) \cdot (\lambda-1) \\ &= -(\lambda-1)(\lambda-4)(\lambda-4) \\ &\text{令 } -(\lambda-1)(\lambda-4)(\lambda-4) = 0 \\ &\Rightarrow \lambda_1 = 1, \lambda_2 = 4, \lambda_3 = 4 \Rightarrow \text{特征值： } \lambda_1 = 1, \lambda_2 = \lambda_3 = 4 \end{aligned}$$

$$\text{对于 } \lambda_1 = 1: \begin{bmatrix} 3-1 & -1 & -1 \\ -1 & 3-1 & -1 \\ -1 & -1 & 3-1 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \Rightarrow \begin{cases} 2x_1 - x_2 - x_3 = 0 \\ -x_1 + 2x_2 - x_3 = 0 \\ -x_1 - x_2 - 2x_3 = 0 \end{cases}$$

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \xrightarrow{\substack{\text{行}_2 + \frac{1}{2} \text{行}_1 \\ \text{行}_3 + \frac{1}{2} \text{行}_1}} \begin{bmatrix} 2 & -1 & -1 \\ 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & -\frac{3}{2} & \frac{3}{2} \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \xrightarrow{\text{行}_3 + \text{行}_2} \begin{bmatrix} 2 & -1 & -1 \\ 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array}$$

$$\xrightarrow{\substack{\text{行}_1 \cdot \frac{1}{2} \\ \text{行}_2 \cdot \frac{2}{3}}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \xrightarrow{\text{行}_1 + \frac{1}{2} \text{行}_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array}$$

$$\Rightarrow \begin{cases} x_1 - x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = 1 \cdot x_3 \\ x_2 = 1 \cdot x_3 \\ x_3 = 1 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ 其中 } k_1 \text{ 为任意常数}$$

$$\Rightarrow \text{特征向量为 } k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} (k_1 \neq 0)$$

$$\text{对于 } \lambda_2 = \lambda_3 = 4: \begin{bmatrix} 3-4 & -1 & -1 \\ -1 & 3-4 & -1 \\ -1 & -1 & 3-4 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \Rightarrow \begin{cases} -x_1 - x_2 - x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \end{cases}$$

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \xrightarrow{\substack{\text{行}_2 - \text{行}_1 \\ \text{行}_3 - \text{行}_1}} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \xrightarrow{\text{行}_1 \cdot (-1)} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} 0 \\ 0 \\ 0 \end{array}$$

$$\Rightarrow \{1x_1 + 1x_2 + 1x_3 = 0\} \Rightarrow \begin{cases} x_1 = -x_2 - x_3 \\ x_2 = x_2 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \cdot x_2 - 1 \cdot x_3 \\ x_2 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_3 = 0 \cdot x_2 + 1 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \text{ 其中 } k_2, k_3 \text{ 为任意常数}$$

$$\Rightarrow \text{特征向量为 } k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} (k_2, k_3 \text{ 不同时为零})$$

(1) 求出特征值与特征向量

(2) 写出 $\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$

(3) 令 $P = \begin{bmatrix} k_1 & k_2 & k_3 \\ \text{后面的} & \text{后面的} & \text{后面的} \\ \text{向量} & \text{向量} & \text{向量} \end{bmatrix}$

练习1. 设矩阵 $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ，求可逆变换矩阵 P 使 $P^{-1}AP = \Lambda$ 为对角矩阵

解： $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{vmatrix} 1 & 2-\lambda & 0 \\ 2-\lambda & 1 & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_2 - (2-\lambda) \cdot \text{行}_1} \begin{vmatrix} 1 & 2-\lambda & 0 \\ 0 & (3-\lambda)(\lambda-1) & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix}$
 $= -[1 \cdot (3-\lambda)(\lambda-1) \cdot (3-\lambda)]$
 $= -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1)$
 令 $-(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1) = 0$
 $\Rightarrow \lambda_1 = 3, \lambda_2 = 3, \lambda_3 = 1 \Rightarrow$ 特征值： $\lambda_1 = \lambda_2 = 3, \lambda_3 = 1$

对于 $\lambda_1 = \lambda_2 = 3$: $\begin{bmatrix} 2-3 & 1 & 0 \\ 1 & 2-3 & 0 \\ 0 & 0 & 3-3 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \Rightarrow \begin{cases} -x_1 + x_2 + 0 \cdot x_3 = 0 \\ x_1 - x_2 + 0 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 0 \end{cases}$
 $\begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \xrightarrow{\text{行}_2 + \text{行}_1} \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \xrightarrow{\text{行}_1 \cdot (-1)} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$
 $\Rightarrow \{1x_1 - 1x_2 + 0x_3 = 0\} \Rightarrow \{x_1 = x_2\} \Rightarrow \begin{cases} x_1 = x_2 \\ x_2 = x_2 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_2 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_3 = 0 \cdot x_2 + 1 \cdot x_3 \end{cases}$
 $\Rightarrow$ 通解为 $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ ，其中 k_1, k_2 为任意常数
 $\Rightarrow$ 特征向量为 $k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ (k_1, k_2 不同时为零)

对于 $\lambda_3 = 1$: $\begin{bmatrix} 2-1 & 1 & 0 \\ 1 & 2-1 & 0 \\ 0 & 0 & 3-1 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \Rightarrow \begin{cases} x_1 + x_2 + 0 \cdot x_3 = 0 \\ x_1 + x_2 + 0 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 2x_3 = 0 \end{cases}$
 $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \xrightarrow{\text{行}_2 - \text{行}_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \xrightarrow{\text{行}_2 \cdot \frac{1}{2}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$
 $\Rightarrow \begin{cases} 1x_1 + 1x_2 + 0x_3 = 0 \\ 0x_1 + 0x_2 + 1x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -x_2 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \cdot x_2 \\ x_2 = 1 \cdot x_2 \\ x_3 = 0 \cdot x_2 \end{cases}$
 $\Rightarrow$ 通解为 $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ ，其中 k_3 为任意常数
 $\Rightarrow$ 特征向量为 $k_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ ($k_3 \neq 0$)

$\Lambda = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

(1) 求出特征值与特征向量

(2) 写出 $\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$

(3) 令 $P = \begin{bmatrix} k_1 & k_2 & k_3 \\ \text{后面的向量} & \text{后面的向量} & \text{后面的向量} \end{bmatrix}$

练习2. 设矩阵 $A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$ ，求可逆变换矩阵 P 使 $P^{-1}AP = \Lambda$ 为对角矩阵

解： $A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} \Rightarrow \begin{vmatrix} 3-\lambda & -1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & -1 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 3-\lambda & -1 & -1 \\ -1 & -1 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_3 - \text{行}_1} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \\ 0 & \lambda-4 & 4-\lambda \end{vmatrix}$

$\xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \end{vmatrix} \xrightarrow{\text{行}_3 - (\lambda-2) \cdot \text{行}_2} \begin{vmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & 0 & (\lambda-4)(\lambda-1) \end{vmatrix}$

$= -1 \cdot (\lambda-4) \cdot (\lambda-4)(\lambda-1)$
 $= -(\lambda-1)(\lambda-4)(\lambda-4)$
 令 $-(\lambda-1)(\lambda-4)(\lambda-4) = 0$
 $\Rightarrow \lambda_1 = 1, \lambda_2 = 4, \lambda_3 = 4 \Rightarrow$ 特征值： $\lambda_1 = 1, \lambda_2 = \lambda_3 = 4$

对于 $\lambda_1 = 1$: $\begin{bmatrix} 3-1 & -1 & -1 & | & 0 \\ -1 & 3-1 & -1 & | & 0 \\ -1 & -1 & 3-1 & | & 0 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 & | & 0 \\ -1 & 2 & -1 & | & 0 \\ -1 & -1 & 2 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} 2x_1 - x_2 - x_3 = 0 \\ -x_1 + 2x_2 - x_3 = 0 \\ -x_1 - x_2 - 2x_3 = 0 \end{cases}$

$\begin{bmatrix} 2 & -1 & -1 & | & 0 \\ -1 & 2 & -1 & | & 0 \\ -1 & -1 & 2 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 + \frac{1}{2} \text{行}_1, \text{行}_3 + \frac{1}{2} \text{行}_1} \begin{bmatrix} 2 & -1 & -1 & | & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} & | & 0 \\ 0 & -\frac{3}{2} & \frac{3}{2} & | & 0 \end{bmatrix} \xrightarrow{\text{行}_3 + \text{行}_2} \begin{bmatrix} 2 & -1 & -1 & | & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$

$\xrightarrow{\text{行}_1 \cdot \frac{1}{2}, \text{行}_2 \cdot \frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_1 + \frac{1}{2} \text{行}_2} \begin{bmatrix} 1 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$

$\Rightarrow \begin{cases} x_1 - x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = 1 \cdot x_3 \\ x_2 = 1 \cdot x_3 \\ x_3 = 1 \cdot x_3 \end{cases}$

$\Rightarrow$ 通解为 $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ，其中 k_1 为任意常数 $\Rightarrow$ 特征向量为 $k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ($k_1 \neq 0$)

对于 $\lambda_2 = \lambda_3 = 4$: $\begin{bmatrix} 3-4 & -1 & -1 & | & 0 \\ -1 & 3-4 & -1 & | & 0 \\ -1 & -1 & 3-4 & | & 0 \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 - x_2 - x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \end{cases}$

$\begin{bmatrix} -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 - \text{行}_1, \text{行}_3 - \text{行}_1} \begin{bmatrix} -1 & -1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_1 \cdot (-1)} \begin{bmatrix} 1 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$

$\Rightarrow \{1x_1 + 1x_2 + 1x_3 = 0\} \Rightarrow \begin{cases} x_1 = -x_2 - x_3 \\ x_2 = x_2 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \cdot x_2 - 1 \cdot x_3 \\ x_2 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_3 = 0 \cdot x_2 + 1 \cdot x_3 \end{cases}$

$\Rightarrow$ 通解为 $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ ，其中 k_2, k_3 为任意常数 $\Rightarrow$ 特征向量为 $k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ (k_2, k_3 不同时为零)

$\Lambda = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

知识点16：求二次型f的矩阵A

B站：猴博士爱讲课
有相关的免费视频

- (1)求出三行三列的矩阵
- (2)将 x_1^2 的系数分配给1行1列
- 将 x_2^2 的系数分配给2行2列
- 将 x_3^2 的系数分配给3行3列
- 将 x_1x_2 的系数对半分，一半给1行2列
- 一半给2行1列
- 将 x_1x_3 的系数对半分，一半给1行3列
- 一半给3行1列
- 将 x_2x_3 的系数对半分，一半给2行3列
- 一半给3行2列

练习1. 设 $f(x_1, x_2, x_3)=2x_1^2+2x_2^2+3x_3^2+2x_1x_2$ ，写出该二次型的矩阵A

解： $f(x_1, x_2, x_3)=2x_1^2+2x_2^2+3x_3^2+2x_1x_2 = 2x_1^2+2x_2^2+3x_3^2+2x_1x_2+0x_1x_3+0x_2x_3 \Rightarrow A=\begin{bmatrix} 2 & \frac{2}{2} & \frac{0}{2} \\ \frac{2}{2} & 2 & \frac{0}{2} \\ \frac{0}{2} & \frac{0}{2} & 3 \end{bmatrix}=\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

练习题2. 求化二次型 $f(x_1, x_2, x_3)=3x_1^2+3x_2^2+3x_3^2-2x_1x_2-2x_1x_3-2x_2x_3$ 的矩阵A

解： $f(x_1, x_2, x_3)=3x_1^2+3x_2^2+3x_3^2-2x_1x_2-2x_1x_3-2x_2x_3 \Rightarrow A=\begin{bmatrix} 3 & \frac{-2}{2} & \frac{-2}{2} \\ \frac{-2}{2} & 3 & \frac{-2}{2} \\ \frac{-2}{2} & \frac{-2}{2} & 3 \end{bmatrix}=\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$

(1) 写出二次型 f 的矩阵 A

(2) 求出矩阵 A 的特征值与特征向量

(3) 求出矩阵 A 的正交矩阵 Q

(4) 标准形： $f = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$

$$\text{正交变换: } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \text{正交矩阵 } Q \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

练习1. 设 $f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2$ 。

(1) 写出该二次型的矩阵 A ;

(2) 求正交矩阵 Q 使得 $Q^T A Q$ 为对角矩阵;

(3) 给出正交变换，化该二次型为标准形。

$$\text{解: (1) } f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 = 2x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 + 0x_1x_3 + 0x_2x_3 \Rightarrow A = \begin{bmatrix} 2 & \frac{2}{2} & \frac{0}{2} \\ \frac{2}{2} & 2 & \frac{0}{2} \\ \frac{0}{2} & \frac{0}{2} & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$(2) A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{vmatrix} 1 & 2-\lambda & 0 \\ 2-\lambda & 1 & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} \xrightarrow{\text{行}_2 - (2-\lambda) \cdot \text{行}_1} \begin{vmatrix} 1 & 2-\lambda & 0 \\ 0 & (3-\lambda)(\lambda-1) & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix}$$

$$= -[1 \cdot (3-\lambda)(\lambda-1) \cdot (3-\lambda)]$$

$$= -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1)$$

$$\text{令 } -(3-\lambda) \cdot (3-\lambda) \cdot (\lambda-1) = 0$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = 3, \lambda_3 = 1 \Rightarrow \text{特征值: } \lambda_1 = \lambda_2 = 3, \lambda_3 = 1$$

$$\text{对于 } \lambda_1 = \lambda_2 = 3: \begin{bmatrix} 2-3 & 1 & 0 & | & 0 \\ 1 & 2-3 & 0 & | & 0 \\ 0 & 0 & 3-3 & | & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 + x_2 + 0 \cdot x_3 = 0 \\ x_1 - x_2 + 0 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 0 \end{cases}$$

$$\begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 + \text{行}_1} \begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_1 \cdot (-1)} \begin{bmatrix} 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\Rightarrow \{1x_1 - 1x_2 + 0x_3 = 0\} \Rightarrow \{x_1 = x_2\} \Rightarrow \begin{cases} x_1 = x_2 \\ x_2 = x_2 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_2 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_3 = 0 \cdot x_2 + 1 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \text{ 其中 } k_1, k_2 \text{ 为任意常数}$$

$$\Rightarrow \text{特征向量为 } k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} k_1 \\ k_1 \\ k_2 \end{bmatrix} \quad (k_1, k_2 \text{ 不同时为零})$$

$$\text{对于 } \lambda_3 = 1: \begin{bmatrix} 2-1 & 1 & 0 & | & 0 \\ 1 & 2-1 & 0 & | & 0 \\ 0 & 0 & 3-1 & | & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 + x_2 + 0 \cdot x_3 = 0 \\ x_1 + x_2 + 0 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 2x_3 = 0 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 - \text{行}_1} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 \cdot \frac{1}{2}} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 1x_1 + 1x_2 + 0x_3 = 0 \\ 0x_1 + 0x_2 + 1x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -x_2 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \cdot x_2 \\ x_2 = 1 \cdot x_2 \\ x_3 = 0 \cdot x_2 \end{cases}$$

$$\Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \text{ 其中 } k_3 \text{ 为任意常数}$$

$$\Rightarrow \text{特征向量为 } k_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -k_3 \\ k_3 \\ 0 \end{bmatrix} \quad (k_3 \neq 0)$$

知识点17：求二次型f的标准型和正交变换

- (1)写出二次型f的矩阵A
- (2)求出矩阵A的特征值与特征向量
- (3)求出矩阵A的正交矩阵Q
- (4)标准形： $f=\lambda_1y_1^2+\lambda_2y_2^2+\lambda_3y_3^2$

正交变换： $\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\text{正交矩阵}Q\cdot\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$

练习1. 设 $f(x_1,x_2,x_3)=2x_1^2+2x_2^2+3x_3^2+2x_1x_2$ 。

- (1) 写出该二次型的矩阵A；
- (2) 求正交矩阵Q使得 $Q^T A Q$ 为对角矩阵；
- (3) 给出正交变换，化该二次型为标准形。

解：(1) $A=\begin{bmatrix}2 & 1 & 0\\1 & 2 & 0\\0 & 0 & 3\end{bmatrix}$

(2)A的特征值： $\lambda_1=\lambda_2=3,\lambda_3=1$

对于 $\lambda_1=\lambda_2=3$ ：特征向量为 $k_1\begin{bmatrix}1\\1\\0\end{bmatrix}+k_2\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}k_1\\k_1\\k_2\end{bmatrix}$ (k_1,k_2 不同时为零)

对于 $\lambda_3=1$ ：特征向量为 $k_3\begin{bmatrix}-1\\1\\0\end{bmatrix}=\begin{bmatrix}-k_3\\k_3\\0\end{bmatrix}$ ($k_3 \neq 0$)

对角矩阵 $A=\begin{bmatrix}3 & 0 & 0\\0 & 3 & 0\\0 & 0 & 1\end{bmatrix}$

$Q_{\text{半成品}}=\begin{bmatrix}k_1 & k_1 & -k_3\\k_1 & k_1 & k_3\\k_2 & k_2 & 0\end{bmatrix}=\begin{bmatrix}1 & k_1 & -1\\1 & k_1 & 1\\0 & 1 & 0\end{bmatrix}$

$\uparrow$
【 $k_1=1$ ，其他=0】

$\uparrow$
【 $k_2=1$ ，其他不动】

$\uparrow$
【 $k_3=1$ ，其他=0】

【∵任意两列都正交 ∴第一列与第二列正交，即 $1\cdot k_1+1\cdot k_1+0\cdot 1=0\Rightarrow k_1=0$ 】

$=\begin{bmatrix}1 & 0 & -1\\1 & 0 & 1\\0 & 1 & 0\end{bmatrix}$

(这是草稿
不写到卷面上)

$Q=\begin{bmatrix}\frac{1}{\sqrt{1^2+1^2+0^2}} & \frac{0}{\sqrt{0^2+0^2+1^2}} & \frac{-1}{\sqrt{(-1)^2+1^2+0^2}}\\\frac{1}{\sqrt{1^2+1^2+0^2}} & \frac{0}{\sqrt{0^2+0^2+1^2}} & \frac{1}{\sqrt{(-1)^2+1^2+0^2}}\\\frac{0}{\sqrt{1^2+1^2+0^2}} & \frac{1}{\sqrt{0^2+0^2+1^2}} & \frac{0}{\sqrt{(-1)^2+1^2+0^2}}\end{bmatrix}=\begin{bmatrix}\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\\0 & 1 & 0\end{bmatrix}$

(3)正交变换为 $\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=Q\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}\Rightarrow\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\\0 & 1 & 0\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$ ，二次型的标准形是 $f=3y_1^2+3y_2^2+1y_3^2=3y_1^2+3y_2^2+y_3^2$

(1) 写出二次型 f 的矩阵 A

(2) 求出矩阵 A 的特征值与特征向量

(3) 求出矩阵 A 的正交矩阵 Q

(4) 标准形: $f = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$

$$\text{正交变换: } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \text{正交矩阵 } Q \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

练习题2. 用正交变换化二次型 $f(x_1, x_2, x_3) = 3x_1^2 + 3x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3$ 为标准形, 并写出相应的正交变换。

$$\text{解: } f(x_1, x_2, x_3) = 3x_1^2 + 3x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3 \Rightarrow A = \begin{bmatrix} 3 & -2 & -2 \\ -2 & 3 & -2 \\ -2 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 3-\lambda & -1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & -1 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_1 \text{与行}_2 \text{互换}} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 3-\lambda & -1 & -1 \\ -1 & -1 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_2 + (3-\lambda) \cdot \text{行}_1} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \\ -1 & -1 & 3-\lambda \end{bmatrix} \xrightarrow{\text{行}_3 - \text{行}_1} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \\ 0 & \lambda-4 & 4-\lambda \end{bmatrix}$$

$$\xrightarrow{\text{行}_2 \text{与行}_3 \text{互换}} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & (\lambda-4)(\lambda-2) & \lambda-4 \end{bmatrix} \xrightarrow{\text{行}_3 - (\lambda-2) \cdot \text{行}_2} \begin{bmatrix} -1 & 3-\lambda & -1 \\ 0 & \lambda-4 & 4-\lambda \\ 0 & 0 & (\lambda-4)(\lambda-1) \end{bmatrix}$$

$$= -1 \cdot (\lambda-4) \cdot (\lambda-4) \cdot (\lambda-1)$$

$$= -(\lambda-1)(\lambda-4)(\lambda-4)$$

$$\text{令 } -(\lambda-1)(\lambda-4)(\lambda-4) = 0$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 4, \lambda_3 = 4 \Rightarrow \text{特征值: } \lambda_1 = 1, \lambda_2 = \lambda_3 = 4$$

$$\text{对于 } \lambda_1 = 1: \begin{bmatrix} 3-1 & -1 & -1 & | & 0 \\ -1 & 3-1 & -1 & | & 0 \\ -1 & -1 & 3-1 & | & 0 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 & | & 0 \\ -1 & 2 & -1 & | & 0 \\ -1 & -1 & 2 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} 2x_1 - x_2 - x_3 = 0 \\ -x_1 + 2x_2 - x_3 = 0 \\ -x_1 - x_2 - 2x_3 = 0 \end{cases}$$

$$\begin{bmatrix} 2 & -1 & -1 & | & 0 \\ -1 & 2 & -1 & | & 0 \\ -1 & -1 & 2 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 + \frac{1}{2} \text{行}_1, \text{行}_3 + \frac{1}{2} \text{行}_1} \begin{bmatrix} 2 & -1 & -1 & | & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} & | & 0 \\ 0 & -\frac{3}{2} & \frac{3}{2} & | & 0 \end{bmatrix} \xrightarrow{\text{行}_3 + \text{行}_2} \begin{bmatrix} 2 & -1 & -1 & | & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\xrightarrow{\text{行}_1 \cdot \frac{1}{2}, \text{行}_2 \cdot \frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_1 + \frac{1}{2} \text{行}_2} \begin{bmatrix} 1 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 - x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = 1 \cdot x_3 \\ x_2 = 1 \cdot x_3 \\ x_3 = 1 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ 其中 } k_1 \text{ 为任意常数} \Rightarrow \text{特征向量为 } k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} k_1 \\ k_1 \\ k_1 \end{bmatrix} (k_1 \neq 0)$$

$$\text{对于 } \lambda_2 = \lambda_3 = 4: \begin{bmatrix} 3-4 & -1 & -1 & | & 0 \\ -1 & 3-4 & -1 & | & 0 \\ -1 & -1 & 3-4 & | & 0 \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 - x_2 - x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \end{cases}$$

$$\begin{bmatrix} -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \\ -1 & -1 & -1 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_2 - \text{行}_1, \text{行}_3 - \text{行}_1} \begin{bmatrix} -1 & -1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{行}_1 \cdot (-1)} \begin{bmatrix} 1 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\Rightarrow \{1x_1 + 1x_2 + 1x_3 = 0\} \Rightarrow \begin{cases} x_1 = -x_2 - x_3 \\ x_2 = x_2 \\ x_3 = x_3 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \cdot x_2 - 1 \cdot x_3 \\ x_2 = 1 \cdot x_2 + 0 \cdot x_3 \\ x_3 = 0 \cdot x_2 + 1 \cdot x_3 \end{cases}$$

$$\Rightarrow \text{通解为 } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \text{ 其中 } k_2, k_3 \text{ 为任意常数} \Rightarrow \text{特征向量为 } k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -k_2 - k_3 \\ k_2 \\ k_3 \end{bmatrix} (k_2, k_3 \text{ 不同时为零})$$

知识点17：求二次型f的标准型和正交变换

- (1)写出二次型f的矩阵A
- (2)求出矩阵A的特征值与特征向量
- (3)求出矩阵A的正交矩阵Q
- (4)标准形： $f=\lambda_1y_1^2+\lambda_2y_2^2+\lambda_3y_3^2$

正交变换： $\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\text{正交矩阵}Q\cdot\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$

练习题2. 用正交变换化二次型 $f(x_1, x_2, x_3)=3x_1^2+3x_2^2+3x_3^2-2x_1x_2-2x_1x_3-2x_2x_3$ 为标准形，并写出相应的正交变换。

解： $A=\begin{bmatrix}3 & -1 & -1\\-1 & 3 & -1\\-1 & -1 & 3\end{bmatrix}$

A的特征值： $\lambda_1=1, \lambda_2=\lambda_3=4$

对于 $\lambda_1=1$ ：特征向量为 $k_1\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}k_1\\k_1\\k_1\end{bmatrix}$ ($k_1 \neq 0$)

对于 $\lambda_2=\lambda_3=4$ ：特征向量为 $k_2\begin{bmatrix}-1\\1\\0\end{bmatrix}+k_3\begin{bmatrix}-1\\0\\1\end{bmatrix}=\begin{bmatrix}-k_2-k_3\\k_2\\k_3\end{bmatrix}$ (k_2, k_3 不同时为零)

对角矩阵 $A=\begin{bmatrix}1 & 0 & 0\\0 & 4 & 0\\0 & 0 & 4\end{bmatrix}$

$Q_{\text{半成品}}=\begin{bmatrix}k_1 & -k_2-k_3 & -k_2-k_3\\k_1 & k_2 & k_2\\k_1 & k_3 & k_3\end{bmatrix}=\begin{bmatrix}1 & -1-k_3 & -1\\1 & 1 & 0\\1 & k_3 & 1\end{bmatrix}$

【 $k_1=1$ ，其他=0】

【 $k_2=1$ ，其他不动】

【 $k_3=1$ ，其他=0】

【 $\because$ 任意两列都正交】

【 $\because$ 第二列与第三列正交】

【即 $(-1-k_3)\cdot(-1)+1\cdot0+k_3\cdot1=0 \Rightarrow k_3=-\frac{1}{2}$ 】

（这是草稿
不写到卷面上）

$$Q=\begin{bmatrix}\frac{1}{\sqrt{1^2+1^2+1^2}} & \frac{-\frac{1}{2}}{\sqrt{(-\frac{1}{2})^2+1^2+(-\frac{1}{2})^2}} & \frac{-1}{\sqrt{(-1)^2+0^2+1^2}}\\\frac{1}{\sqrt{1^2+1^2+1^2}} & \frac{1}{\sqrt{(-\frac{1}{2})^2+1^2+(-\frac{1}{2})^2}} & \frac{0}{\sqrt{(-1)^2+0^2+1^2}}\\\frac{1}{\sqrt{1^2+1^2+1^2}} & \frac{-\frac{1}{2}}{\sqrt{(-\frac{1}{2})^2+1^2+(-\frac{1}{2})^2}} & \frac{1}{\sqrt{(-1)^2+0^2+1^2}}\end{bmatrix}=\begin{bmatrix}\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0\\\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}}\end{bmatrix}$$

正交变换为 $\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=Q\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}\Rightarrow\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0\\\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$ ，二次型的标准形是 $f=1y_1^2+4y_2^2+4y_3^2=y_1^2+4y_2^2+4y_3^2$